

Congestion-Aware Joint Charging-Station Placement, Capacity, and Swarm-UAV Trajectory Design for Age-Optimal Data Collection

Phuc Hao Do, Tran Duc Le, Truong Duy Dinh, and Van Dai Pham

Abstract—Unmanned aerial vehicle (UAV) swarms can keep Internet of Things data fresh, but energy-limited UAVs must recharge at a small number of shared charging stations, which then become a contended resource. We study the joint design of charging-station placement, port capacity, swarm trajectory, and UAV-to-station assignment to minimize the network peak Age of Information (AoI). A key observation is that the charging system is a closed population of only M UAVs, so it must be modeled as a finite-source (machine-repair) queue rather than an open-arrival queue. A discrete-event simulation confirms the finite-source model within about 4%, whereas the open queue overpredicts the waiting time by roughly $6\times$ and distorts the design decisions. We prove two structural results: a congestion threshold that makes the peak AoI non-monotone in the swarm size, with a provisioning law linking the optimal swarm size to the charging capacity, and a placement inversion showing that the coverage-optimal placement is AoI-suboptimal under contention. We then develop a block-coordinate solver with a GPU close-enough traveling-salesman trajectory optimizer. Across multi-seed experiments, the proposed method reduces the peak AoI by up to 12.7% over a strong single-station baseline, with each component isolated by an ablation and confirmed by a Wilcoxon signed-rank test. We also delineate the regime in which a single pooled station is already competitive.

Index Terms—Age of Information, unmanned aerial vehicles, UAV swarm, data collection, charging-station placement, finite-source queue, trajectory optimization, Internet of Things.

I. INTRODUCTION

Many Internet of Things applications depend on data that is not only correct but also fresh. In precision agriculture, environmental monitoring, disaster response, and industrial inspection, a control loop or an operator acts on the most recent sensor reading, so a stale reading can be worse than no reading at all. Ground sensors in these settings are often sparse, spread over a wide area, and far from any gateway,

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The source code and data that reproduce all results are available at <https://github.com/haodpsut/swarm-uav-aoi-congestion>.

which makes continuous fresh collection difficult with fixed infrastructure alone.

Unmanned aerial vehicles (UAVs) are attractive data collectors for such deployments: they fly close to ground sensors, establish strong line-of-sight links, and reach areas without fixed infrastructure. When the collected data is time-sensitive, the relevant metric is the Age of Information (AoI) [1], that is, the time elapsed since the freshest delivered update was generated. Keeping the network AoI low requires a UAV to revisit each sensor often, which conflicts with the limited battery of a rotary-wing UAV: the platform must periodically leave the field to recharge at a charging station.

A single UAV can only keep a small area fresh, so practical deployments use a *swarm*. A swarm sharing a small number of charging stations creates a problem that a single UAV never faces: the stations become a *contended* resource. When several UAVs need to recharge at the same time, they compete for a limited number of charging ports and must wait, and this waiting time inflates the AoI. The system designer must therefore decide jointly where to place the charging stations, how many ports to install at each, how the swarm should fly, and which UAV recharges where.

A. Limitations of prior work

Existing studies address parts of this problem but not the whole. The closest work jointly optimizes charging-station placement and trajectory, but for a *single* UAV, so there is no contention [2]. Multi-UAV studies that model charging either assume a single fixed facility [3], use mid-flight laser charging with fixed emitters and no queue [4], or use mobile ground vehicles that follow the UAVs and thereby remove the fixed-station contention [5], [6]. Swarm-AoI studies that ignore energy altogether [7] cannot capture recharging at all. A separate line models UAV charging stations as a queue with waiting time [8], but without any AoI objective, trajectory, or placement. A naive queueing model is misleading here: the charging system is a *closed* population of only M UAVs, so an open $M/M/c$ queue substantially overpredicts the wait, as we show against a discrete-event simulation.

B. Contributions

- To the best of our knowledge, we formulate the first joint charging-station *placement*, port *capacity*, swarm *trajectory*, and *assignment* problem for peak-AoI fresh data collection with a contended, multi-server charging

resource, as a mixed-integer nonlinear program (Section III).

- We model the charging stations as a *finite-source* (machine-repair) queue and validate it against a discrete-event simulation. The finite-source model matches the simulation within about 4% and is a conservative upper bound on the real wait, whereas the open $M/M/c$ overpredicts by about $6\times$ and distorts the design decisions.
- We prove two structural results (Section IV): a *congestion threshold* that makes the AoI non-monotone in the swarm size, with a provisioning law linking the optimal swarm size to the charging capacity, and a *placement inversion* showing that the coverage-optimal placement is AoI-suboptimal under contention.
- We design a block-coordinate solver with a GPU close-enough-TSP trajectory optimizer (Section V) and evaluate it against single-UAV, single-station, coverage-optimal, and round-robin baselines (Section VI). The proposed method reduces peak AoI significantly, and we also identify the regime in which a single pooled station is already competitive. We further show that the solver attains the *conditional combinatorial optimum* exactly for $M \leq 6$ and within 0.11% at $M = 8$, and that its advantage is robust to the collection radius, the charge time, and non-uniform sensor distributions.

The remainder of the paper is organized as follows. Section II reviews related work, Section III presents the model and problem, Section IV the structural analysis, Section V the solver, and Section VI the experiments. Section VII concludes.

II. RELATED WORK

A. Age of Information foundations

AoI was introduced in [1] as a destination-centric measure of data freshness, capturing the time elapsed since the generation of the most recently received update. The concept has since matured into a substantial body of theory, surveyed comprehensively in [9] and developed as a self-contained metric and analysis tool in [10]. Two threads of this literature are directly relevant to our problem. The first concerns sampling and transmission policy: [11] shows that, contrary to intuition, freshness is not always improved by updating as often as possible, and that an optimal policy may deliberately wait, while [12] analyzes how buffer and packet management shape the age process. The second concerns scheduling across contending sources: [13] derives age-optimal and near-optimal policies for broadcasting updates from many sources over a shared wireless channel. These foundations motivate our use of peak AoI as the design objective and, more subtly, they anticipate the tension our formulation confronts, namely that serving more sources with a shared resource (here, a shared charger) can raise age once contention is taken into account.

B. UAV communications, trajectory, and placement

Unmanned aerial vehicles have become a central platform for flexible wireless service, with the opportunities and open problems laid out in [14] and a broad tutorial treatment in [15]; the networking of cooperating aerial nodes is surveyed in [16].

Trajectory design is the dominant control lever in this setting: [17] jointly optimizes the paths and communication of multiple UAVs, and for data collection specifically [18] minimizes collection energy while [19] minimizes the flight time needed to gather data from a field of sensors. The quality of the air-to-ground link that these designs exploit is governed by the elevation-dependent line-of-sight probability characterized in [20] and by the aerial channel models standardized in [21], which we draw on when specifying the collection link. Two combinatorial structures recur in our own design. Deciding where to site a small number of charging stations is a facility-location problem, whose models and algorithms are treated in [22]. Planning a visiting tour in which a source is served whenever the UAV enters its communication range is a close-enough traveling salesman problem, formulated and solved exactly in [23] and heuristically in [24]. Our formulation couples these two structures, which are typically studied in isolation, to the freshness and charging dynamics described below.

C. AoI-oriented UAV data collection

Building on these foundations, AoI has become a standard objective for UAV-assisted data collection, where a UAV visits distributed sources and the design goal is to keep the collected information fresh. Single-UAV formulations establish the basic tradeoffs: [25] designs UAV-aided collection to maximize information freshness across a sensor field, and [26] minimizes average peak AoI in a UAV-assisted IoT network, jointly treating trajectory and update scheduling. In the swarm setting, [7] minimizes AoI across multiple cooperating UAVs using multi-agent reinforcement learning, demonstrating that coordinated trajectories can reduce peak AoI relative to uncoordinated flight; learning-based control of multiple UAVs is itself an active area, with resource allocation studied in [27] and deployment and movement design in [28]. That family of formulations, however, assumes flight endurance is unconstrained: it carries no energy state and no recharging model, so the depletion and replenishment dynamics that dominate long missions are absent. Closest to our objective is [2], which jointly optimizes charging station placement, UAV trajectory, and peak AoI, and which we adopt as our point of departure. That work is nonetheless single-UAV: with one vehicle there is no inter-UAV contention for a charger, no need for multi-port capacity, and no swarm assignment, precisely the coupling that our design targets.

D. Energy-limited UAVs and charging

A second line of work makes energy replenishment explicit. The battery lifetime of aerial nodes is a first-order operational constraint, as argued in [29], and wireless power transfer offers one replenishment mechanism whose trajectory and energy tradeoffs are optimized in [30]. Turning to shared charging infrastructure, [3] lets multiple UAVs share a limited charging facility and folds the resulting charge waiting time into the AoI objective, then optimizes it by deep reinforcement learning. The facility there is single and fixed in location, which effectively reduces contention to a single-server queue

and leaves station placement out of scope. Laser-based (LBD) charging is studied in [4], where charging sites are at fixed locations and are treated without queue contention, so neither where to place stations nor how many vehicles may wait is optimized. A different tactic uses mobile chargers: [5] and [6] employ ground vehicles that follow the UAVs to deliver energy on the move. Because the charger comes to the UAV, this removes the fixed-station contention that is central to our setting and does not address the placement of fixed, multi-port stations. Finally, [8] does model UAV charging stations as a contended queue and provides a waiting-time analysis, but it optimizes neither AoI nor trajectory nor station placement, so the queueing model is not coupled to any freshness or mobility decision. Beyond aerial platforms, a broad literature schedules the charging of vehicle and sensor fleets: auction-based time-slot allocation coordinates recharging across a multi-drone network in [31], decentralized protocols shift electric-vehicle charging in time to flatten aggregate demand in [32], and a mobile wireless charger is routed to replenish a sensor fleet in [33]. These dynamic-fleet-charging methods decide when and where each vehicle charges, yet they neither jointly place and size the shared stations nor couple the induced charging wait into an Age-of-Information objective for a UAV swarm, which is the focus of our formulation. When we make contention explicit, we model a station as a finite-source (closed) service system rather than an open queue. This places our analysis in the lineage of the machine-repair, or machine-interference, problem surveyed in [34] and treated within the classical queueing framework of [35], from which we adapt the finite-source multi-server results used in Section III.

E. Charging-station placement and recharge-aware swarm operation

One body of work places or sizes UAV recharging infrastructure, another optimizes AoI for swarms that recharge, and the two are largely disjoint. We discuss the closest works individually; Table I compares them.

Won [36] shows that siting battery-swap stations is NP-hard and solves it jointly with UAV trajectories, but each station is a single swap point with no parallel servers, no waiting, and delivery performance as the objective. Santin *et al.* [37] locate a minimal number of recharge stations while routing a heterogeneous fleet for complete coverage, again without server capacity or queueing and with coverage time as the objective. Chen *et al.* [38] come closest on the infrastructure side, optimizing both *siting and sizing* together with routing, task allocation and recharge scheduling; but the objective is cost and task time, contention is handled by a schedule rather than a queueing model, and freshness plays no role.

On the freshness side, Eldeeb *et al.* [39] minimize AoI and energy for multiple UAV relays and account for flights to recharging depots, but the depots are given, not placed or sized. Shi *et al.* [40] and Lee *et al.* [41] likewise minimize AoI for swarms that recharge at stations, using multi-agent reinforcement learning; in both, the station set is fixed and its capacity is not a decision variable. Zhang *et al.* [42] alone makes charger *scarcity* explicit, with a limited number of piles

TABLE I
CLOSEST RELATED WORK. EVERY LISTED WORK CONSIDERS *multiple* UAVS THAT *self-recharge* AT *fixed* GROUND STATIONS AND OPTIMIZES *trajectories*, SO THOSE FOUR DIMENSIONS ARE OMITTED AS NON-DISCRIMINATING; THE TABLE COMPARES THE DIMENSIONS ON WHICH THE WORKS DIFFER. ✓ MEANS MODELLED AND OPTIMIZED, “FIX” MEANS PRESENT BUT AN INPUT, “—” ABSENT.

Work	Objective	Place- ment	Port sizing	Explicit waiting	Assign- ment
[36]	delivery perf.	✓	—	—	—
[37]	coverage time	✓	—	—	—
[39]	AoI + energy	fix	—	—	—
[42]	travel time	fix	fix	—	—
[38]	cost, task time	✓	✓	—	✓
[41]	AoI	fix	—	—	—
[40]	AoI	fix	—	—	—
This work	peak AoI	✓	✓	✓	✓

per station, yet the pile count is an input, the objective is travel time, and the contention it induces has no waiting-time model.

The pattern is consistent: placement and sizing are studied without freshness, and freshness with the infrastructure taken as given. Missing in all of them is what this paper adds, an *explicit waiting model* for a small closed population sharing a few parallel ports, which is what couples placement and capacity to the AoI.

F. Positioning and gap

Read together, these works each supply a subset of the ingredients our problem requires but never their combination. The same split appears outside Table I: swarm AoI without charging [7], placement with AoI but a single UAV [2], charge waiting at one fixed server without placement [3], and a contended charging queue without AoI, trajectory or placement [8]. We close this gap by jointly optimizing placement, port capacity, swarm trajectory, and assignment under a contended, multi-server, finite-source (closed) charging queue validated against discrete-event simulation. The finite-source aspect is essential, not incidental: with a small closed population competing for a station, an open $M/M/c$ model overpredicts the wait and distorts both placement and capacity decisions, an effect we quantify and correct in Section III. This coupled formulation also yields two structural findings that a decoupled treatment cannot expose: a congestion threshold, beyond which peak AoI is non-monotone in swarm size, and a placement inversion, whereby coverage-optimal station placement becomes AoI-suboptimal once contention is accounted for.

III. SYSTEM MODEL AND PROBLEM FORMULATION

The main symbols are summarized in Table II.

A. Network model

We consider a swarm data-collection system, illustrated in Fig. 1. A set $\mathcal{K} = \{1, \dots, K\}$ of ground sensors is deployed over a square field, with sensor k located at $\mathbf{w}_k \in \mathbb{R}^2$. A swarm of M rotary-wing UAVs, indexed by $\mathcal{M} = \{1, \dots, M\}$, flies at a fixed altitude H with horizontal position $\mathbf{q}_m(t) \in \mathbb{R}^2$

TABLE II
SUMMARY OF NOTATION.

Symbol	Meaning
$K, \mathcal{K}$	number and set of ground sensors
M	swarm size (number of UAVs)
$\mathcal{S}, S_{\max}$	candidate sites, station budget
$x_s \in \{0, 1\}$	open a station at candidate site s
c_s, C_{tot}	ports at station s , total port budget
$\mathbf{q}_m(t)$	trajectory of UAV m
a_m	station assigned to UAV m
$A_k(t), \bar{A}$	AoI of sensor k , network peak AoI (objective)
V, H	cruise speed, altitude
$E_{\max}, \eta$	battery, reserve fraction
$\tau_{\text{fly}}, \tau_{\text{charge}}$	flight budget, charge time
T_{up}	operating time between charges (incl. travel)
N_s	UAVs assigned to station s
λ, μ	charge-request rate, service rate
W_q, t_{loop}	mean queue wait, sub-field revisit period

over discrete slots $t = 1, \dots, T$ of length δ_t . Each UAV holds a battery with state of charge $e_m(t) \in [0, E_{\max}]$. A set $\mathcal{S} = \{1, \dots, S\}$ of candidate charging-station sites is given, with site s at $\mathbf{b}_s \in \mathbb{R}^2$; the system may build a station and provision it with charging ports.

Freshness is measured by the AoI [1]. Let $u_k(t)$ be the generation time of the freshest update from sensor k that has been delivered by time t . The instantaneous AoI is

$$\text{AoI}_k(t) = t - u_k(t), \quad (1)$$

which grows at unit rate and drops on each successful collection: when a UAV collects at time τ a packet generated at g , u_k jumps to g and $\text{AoI}_k(\tau^+) = \tau - g$. Following the base setting, the network objective is the peak AoI across sensors, let $t_{k,1} < t_{k,2} < \dots$ be the epochs at which an update from sensor k is delivered, so that $\text{AoI}_k(t_{k,i}^-)$ is the age reached just before the i -th delivery, i.e. the *peak* of cycle i . The objective is the expected peak age of the worst sensor,

$$\bar{A} = \max_{k \in \mathcal{K}} \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \text{AoI}_k(t_{k,i}^-), \quad (2)$$

which we seek to minimize.

Three points about this definition, since an earlier version of this work was inconsistent about it. First, (2) is an *expectation over cycles*, not a sample-path worst case: the earlier form $\max_k \limsup_t \text{AoI}_k(t)$ is withdrawn, and (12) below optimizes (2) rather than restating a different functional. Second, (2) is the average of the *peaks*, not the time-average of the age; those differ by a factor close to two, and we measure rather than assert it. Across the 9 configurations of Section VI-H the ratio of the expected peak to the time-average age is 2.00 to 2.00, because the age sawtooths from zero to the revisit interval and the cycle length varies little (coefficient of variation at most 0.040). Reporting a time-average under the name “peak AoI” would therefore halve every number in this paper. Third, using the *mean* charging wait W_q inside (2) is legitimate precisely because (2) is itself an expectation; it would not be legitimate under the withdrawn sample-path form, and Section VI-H quantifies that gap: in the well-provisioned regime this paper recommends, the largest

observed wait exceeds the mean by up to $82.9\times$. A UAV collects from sensor k while $\|\mathbf{q}_m(t) - \mathbf{w}_k\| \leq r_c$.

B. Rotary-wing energy

The charging cadence is not assumed constant: it emerges from how fast a UAV drains its battery in flight. For a rotary-wing UAV cruising at horizontal speed V , the propulsion power follows the closed-form model of Zeng *et al.* [43],

$$P(V) = P_0 \left(1 + \frac{3V^2}{U_{\text{tip}}^2} \right) + P_i \left(\sqrt{1 + \frac{V^4}{4v_0^4}} - \frac{V^2}{2v_0^2} \right)^{1/2} + \frac{1}{2} d_0 \rho s A V^3, \quad (3)$$

where P_0 and P_i are the blade-profile and induced power constants, U_{tip} the rotor tip speed, v_0 the mean rotor induced velocity in hover, and d_0, ρ, s, A the fuselage drag ratio, air density, rotor solidity, and disc area. Hovering during collection consumes $P(0) = P_0 + P_i$. The charge evolves as $e_m(t+1) = e_m(t) - P(V_m(t)) \delta_t$ while flying or collecting, and increases while docked at a charging port.

Only a usable fraction of the pack is spent between charges, keeping a reserve η , so $E_{\text{use}} = (1 - \eta) E_{\max}$. Given a patrol loop of flight time $t_{\text{fly}}^{\text{loop}}$ and hover-collection time $t_{\text{col}}^{\text{loop}}$, the energy drawn per loop is $e_{\text{loop}} = P(V) t_{\text{fly}}^{\text{loop}} + P(0) t_{\text{col}}^{\text{loop}}$, so a UAV can fly $n_{\text{loop}} = \lfloor E_{\text{use}} / e_{\text{loop}} \rfloor$ loops before recharging, giving a physical flight budget between charges

$$\tau_{\text{fly}} = n_{\text{loop}} t_{\text{loop}}, \quad t_{\text{loop}} = t_{\text{fly}}^{\text{loop}} + t_{\text{col}}^{\text{loop}}. \quad (4)$$

A recharge-feasibility condition requires each UAV to retain enough charge to reach its assigned open station at all times: $e_m(t) \geq E_{\text{reach}}(\mathbf{q}_m(t), \mathbf{b}_{a_m})$.

C. Air-to-ground collection

Collection dwell is derived from a physical link rather than assumed instantaneous. With a line-of-sight, free-space channel of reference gain β_0 at 1 m, a sensor at horizontal distance d is seen at slant distance $\sqrt{H^2 + d^2}$, and the achievable rate is

$$R(d) = B \log_2 \left(1 + \frac{p_{\text{tx}} \beta_0}{(H^2 + d^2) N_0 B} \right), \quad (5)$$

with bandwidth B , sensor transmit power p_{tx} , and noise power spectral density N_0 . Collecting a status blob of L bits at the closest-approach distance therefore takes a dwell time $t_{\text{col}}(d) = L/R(d)$, which enters t_{loop} in (4) and hence the revisit period.

D. Finite-source charging queue

The load-bearing coupling of the swarm is contention for charging ports. A station s serving the N_s UAVs assigned to it with c_s parallel ports is *not* an open queue: only N_s UAVs exist, so the request rate falls as more of them are already at the station. The correct model is the finite-source (machine-repair) queue $M/M/c//N_s$ [35].

Each assigned UAV, while patrolling, requests a charge after a mean operating time, at rate $\lambda = 1/T_{\text{up}}$; each port serves at $\mu = 1/\tau_{\text{charge}}$. The operating time between two charge

requests is the whole time a UAV spends away from the station in a cycle, so it must include the round-trip travel to the station and not the flight budget alone:

$$T_{\text{up}} = \tau_{\text{fly}} + \frac{2d(m, s)}{V}, \quad (6)$$

where $d(m, s) = \|\mathbf{q}_m - \mathbf{b}_s\|$. Omitting the travel term overstates λ and overpredicts the wait by roughly a factor of two.

Delivery semantics. Two reviewers asked us to state when a collected update counts as *delivered*, since the age is defined by delivery while the system model describes collection. We assume an ideal backhaul: the instant a UAV collects an update it is available at the destination, so collection time equals delivery time and no backhaul or data-carrying delay enters the age. This is an idealization, appropriate only when a persistent air-to-ground or satellite backhaul is present. A deliver-on-return model, in which the UAV carries the data to a ground node, would add the return flight to every peak and would change (11); the placement and capacity structure we study would survive, but the absolute ages would not. We state the assumption here rather than leave it implicit.

Let $n \in \{0, \dots, N_s\}$ be the number of UAVs at the station. The birth-death chain has state-dependent rates

$$n \rightarrow n+1 : (N_s - n)\lambda, \quad n \rightarrow n-1 : \min(n, c_s)\mu, \quad (7)$$

where the up-rate $(N_s - n)\lambda$ vanishes as $n \rightarrow N_s$: this closed-population term is exactly why the queue cannot blow up. The stationary distribution is

$$p_n = p_0 \prod_{k=0}^{n-1} \frac{(N_s - k)\lambda}{\min(k+1, c_s)\mu}, \quad \sum_{n=0}^{N_s} p_n = 1. \quad (8)$$

The mean number waiting, the effective throughput, and, by Little's law, the mean queue wait are

$$\begin{aligned} L_q &= \sum_{n=0}^{N_s} \max(0, n - c_s) p_n, \\ \lambda_{\text{eff}} &= \sum_{n=0}^{N_s} (N_s - n) \lambda p_n, \\ W_q(N_s, c_s) &= \frac{L_q}{\lambda_{\text{eff}}}. \end{aligned} \quad (9) \quad (10)$$

Because the population is finite, no stability constraint (of the form $\rho_s < 1$) is needed: the wait *saturates* rather than diverging. This finite-source model replaces the open $M/M/c$ (Erlang-C) form, which treats arrivals as an inexhaustible stream and, for the cycling swarm, overpredicts the wait by roughly a factor of six. A discrete-event validation of the true cycling UAVs, reported in Section VI, confirms that (10) tracks the simulation to within a few percent and is a conservative upper bound on the real, near-deterministic wait.

E. Peak-AoI expression

For a UAV m that patrols a sub-field and leaves it to recharge, the expected revisit interval seen by its sensors is one patrol period plus one charging excursion, and the age

of a sensor peaks at exactly that interval just before the next delivery:

$$\text{AoI}_m = t_{\text{loop},m} + \frac{2d(m, s)}{V} + W_q(N_s, c_s) + \tau_{\text{charge}}. \quad (11)$$

The first term is the patrol revisit period, the second the round trip to the assigned station, the third the finite-source queue wait from (10) (the term that couples the swarm), and the last the charging time. Because every term is a mean, (11) is the expected cycle peak of (2) for the sensors served by UAV m ; the network objective (2) is the maximum of (11) over the swarm. This is the quantity Algorithm 1 minimizes, and it is the same functional that (12) states. The coupling in W_q is the mechanism behind the non-monotone behavior of $\bar{A}$ in the swarm size M : more UAVs shrink the coverage term but lengthen the wait, up to the ceiling set by the finite population.

F. Joint optimization problem

We jointly optimize the placement decisions $x_s \in \{0, 1\}$ (build a station at s), the capacities $c_s \in \mathbb{Z}_{\geq 0}$ (ports per station), the UAV trajectories $\{\mathbf{q}_m(\cdot)\}$, and the station assignment $\{a_m\}$. The problem is

$$(P0) \quad \arg \min_{\substack{\mathbf{x}, \mathbf{c}, \\ \{\mathbf{q}_m\}, \{a_m\}}} \bar{A} \text{ as defined in (2)} \quad (12a)$$

$$\text{s.t.} \quad e_m(t) \in [0, E_{\text{max}}], \quad \forall m, t, \quad (12b)$$

$$e_m(t) \geq E_{\text{reach}}(\mathbf{q}_m(t), \mathbf{b}_{a_m}), \quad \forall m, t, \quad (12c)$$

$$\text{AoI}_k(t) \text{ evolves as in (1), } \forall k, t, \quad (12d)$$

$$\sum_s x_s \leq S_{\text{max}}, \quad \sum_s c_s \leq C_{\text{tot}}, \quad (12e)$$

$$c_s \leq C_{\text{port}} x_s, \quad x_s \in \{0, 1\}, \quad c_s \in \mathbb{Z}_{\geq 0}, \quad (12f)$$

$$\|\mathbf{q}_m(t+1) - \mathbf{q}_m(t)\| \leq V_{\text{max}} \delta t, \quad \forall m, t, \quad (12g)$$

$$a_m \in \{s : x_s = 1\}, \quad \forall m, \quad (12h)$$

$$\|\mathbf{q}_m(t) - \mathbf{q}_{m'}(t)\| \geq d_{\text{min}}, \quad \forall m \neq m', t. \quad (12i)$$

Constraint (12c) is recharge feasibility, (12d) ties peak AoI to collection events through (11) and the queue wait (10), (12e) and (12f) enforce the station and port budgets, (12g) is UAV kinematics, (12h) restricts assignment to open stations, and (12i) is inter-UAV separation.

Problem (P0) mixes binary placement $\mathbf{x}$, integer capacity $\mathbf{c}$, continuous trajectories, and a combinatorial assignment, and its objective embeds the nonconvex finite-source wait (10) through the coupling (11). It is therefore a mixed-integer nonlinear program (MINLP) and NP-hard, which motivates the block-coordinate decomposition (trajectory, queue evaluation, capacity allocation, assignment, and outer placement) detailed in Section V.

IV. STRUCTURAL ANALYSIS

We now establish the two structural properties that separate the swarm regime from the single-UAV base case: the peak AoI is non-monotone in the swarm size (Proposition 1),

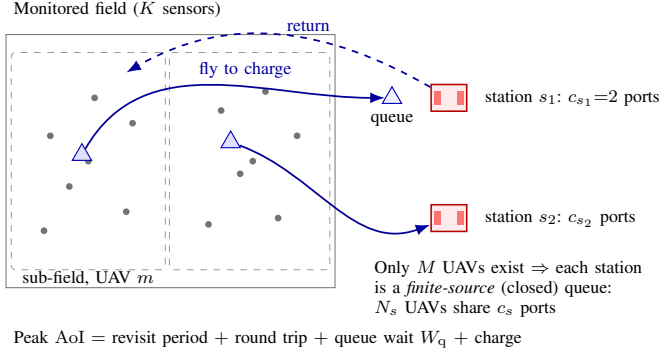


Fig. 1. System model: a UAV swarm patrols sub-fields to keep sensor data fresh, and recharges at a small number of shared, multi-port charging stations. Because only M UAVs exist, the charging stations form a *finite-source* (closed) queue.

and the AoI-optimal station placement inverts from coverage-driven to traffic-driven once the swarm contends for ports (Proposition 2). Throughout, we work with the per-UAV peak AoI of the homogeneous single-station cell derived in Section III,

$$\text{AoI}(M) = \underbrace{t_{\text{loop}}(M)}_{c_1(M)} + \underbrace{W_q(M, c)}_{\text{contention}} + \underbrace{\frac{2 \text{dist}}{V} + \tau_{\text{charge}}}_{c_0}, \quad (13)$$

where M UAVs share one field and one station of c ports, $t_{\text{loop}}(M)$ is the patrol (revisit) time per UAV, $W_q(M, c)$ is the mean charging wait of the finite-source queue $M/M/c/M$ of Section III (population M , per-source operating rate $\lambda = 1/T_{\text{up}}$, port rate $\mu = 1/\tau_{\text{charge}}$), and c_0 collects the recharge round trip and the charging time, which do not depend on M .

A. Congestion threshold: AoI is non-monotone in swarm size

We first isolate the two competing forces in (13): the coverage term $c_1(M)$ that a larger swarm improves, and the wait $W_q(M, c)$ that a larger swarm degrades.

Assumption 1 (Coverage scaling): The patrol term $c_1(M) = t_{\text{loop}}(M)$ is positive, strictly decreasing, and convex in M , with $c_1(M) = \Theta(1/M)$.

The scaling is the standard sub-tour argument: M UAVs partition K sensors of a field of area $\mathcal{A}$ into M sub-fields of area $\mathcal{A}/M$ and K/M sensors each. The length of a near-optimal tour through n points uniformly spread over an area $\mathcal{A}'$ scales as $\Theta(\sqrt{\mathcal{A}'n})$, so each sub-tour has length $\Theta(\sqrt{(\mathcal{A}/M)(K/M)}) = \Theta(\sqrt{\mathcal{A}K}/M)$, giving $t_{\text{loop}}(M) = \Theta(1/M)$. Convexity and monotonicity of $1/M$ carry over to the leading term, and we take them as the modeling assumption above.

Lemma 1 (Monotone, saturating charging wait): In the finite-source queue $M/M/c/M$, the mean wait $W_q(M, c)$ is non-decreasing in the population M , is strictly increasing once $M > c$, and satisfies the bound

$$0 \leq W_q(M, c) \leq \frac{M - c}{c\mu}, \quad M > c. \quad (14)$$

Proof: Let $\{n_t\}$ denote the birth-death chain of Section III on states $n \in \{0, \dots, M\}$ (UAVs at the station), with up-rates $(M - n)\lambda$ and down-rates $\min(n, c)\mu$ [35]. Increasing

the population from M to $M + 1$ adds one source: at every state the up-rate weakly increases (from $(M - n)\lambda$ to $(M + 1 - n)\lambda$) while the down-rate is unchanged. A standard monotone coupling of the two chains keeps the $(M + 1)$ -chain pathwise at least as high as the M -chain, so the stationary number in system is stochastically increasing in M .

We convert this into a statement about the wait through the arriving-UAV view, rather than through the queue length alone: because the effective throughput $\lambda_{\text{eff}}(M)$ also grows with M , monotonicity of the ratio $W_q = L_q/\lambda_{\text{eff}}$ does not follow from monotonicity of L_q by itself. By the arrival theorem for finite-source (machine-repair) queues [35], a UAV that requests a charge in the M -population system finds the station distributed as the *stationary* distribution of the $(M - 1)$ -population system. A UAV that finds n present is served after $(n - c + 1)^+$ successive departures, each exponential of rate $c\mu$ while the c servers stay busy, so its conditional mean wait is $(n - c + 1)^+/(c\mu)$, which is non-decreasing in n . Averaging a non-decreasing function against a distribution that is stochastically increasing in M , $W_q(M, c)$ is non-decreasing in M (and equals the Little's-law form L_q/λ_{eff} evaluated in Section III). The increase is strict once $M > c$, because states $n > c$ then carry positive probability in the arriving-UAV distribution and contribute a strictly positive wait. For (14), an arriving UAV finds at most $M - 1$ others present, so $n \leq M - 1$ and its conditional wait is at most $(M - 1 - c + 1)/(c\mu) = (M - c)/(c\mu)$; taking the expectation gives $W_q(M, c) \leq (M - c)/(c\mu)$. ■

Two features of Lemma 1 drive the result. Below c the wait is essentially zero (no UAV ever queues), and above c it climbs, ultimately at the linear rate $\Theta(M/(c\mu))$ of (14): its forward differences $\Delta W_q(M) \triangleq W_q(M + 1, c) - W_q(M, c)$ are non-negative and rise from $\Delta W_q \approx 0$ in the underloaded region toward the saturation slope $1/(c\mu)$. We take this saturating shape as an explicit assumption.

Assumption 2 (Saturating charging increments): For each fixed c , the increment $\Delta W_q(M, c)$ is non-decreasing in M (so $W_q(\cdot, c)$ is discretely convex), and $\Delta W_q(M, c)$ is non-increasing in c .

Both parts reflect the same mechanism: added contention (larger M) steepens the wait, while added ports (larger c) flatten it and defer saturation. They are verified numerically across the tested field and capacity configurations in Section VI.

Proposition 1 (Congestion threshold and provisioning law): Under Assumptions 1 and 2 and Lemma 1, the peak AoI $\text{AoI}(M)$ in (13) is discretely convex in the swarm size M , and its set of minimizers

$$\mathcal{M}^* = \arg \min_{M \in \{1, 2, \dots\}} \text{AoI}(M), \quad M^* \triangleq \min \mathcal{M}^*, \quad (15)$$

is non-empty, finite, and a block of consecutive integers. Adding UAVs strictly reduces AoI for $M < M^*$ (coverage dominates) and strictly increases it for $M > \max \mathcal{M}^*$ (charging congestion dominates). The minimizer is unique, $\mathcal{M}^* = \{M^*\}$, if and only if $\Delta \text{AoI}(M) \neq 0$ for every M ; a tie requires the coverage gain and the congestion penalty to cancel *exactly* at the crossing index. Moreover M^* is non-

decreasing in the port capacity c : enlarging a station never lowers the optimal swarm size.

Proof: Since c_0 is constant in (13), work with the forward difference

$$\Delta \text{AoI}(M) = \Delta c_1(M) + \Delta W_q(M), \quad (16)$$

where $\Delta c_1(M) = c_1(M+1) - c_1(M)$. By Assumption 1, c_1 is strictly decreasing and convex, so $\Delta c_1(M) < 0$ and $\Delta c_1(M)$ is non-decreasing (it rises toward 0 as $M \rightarrow \infty$). By Lemma 1 $\Delta W_q(M) \geq 0$, and by Assumption 2 it is non-decreasing. Hence the right-hand side of (16) is a sum of two non-decreasing terms, so $\Delta \text{AoI}(M)$ is non-decreasing in M : the sequence $\text{AoI}(M)$ is discretely convex.

It remains to fix the sign of ΔAoI at the two ends. For small M (in particular $M \leq c$) the queue is empty, $\Delta W_q(M) = 0$, while $\Delta c_1(M) < 0$, so $\Delta \text{AoI}(M) < 0$: AoI is still falling.

For large M we do not need the exact saturation slope, only a strictly positive floor on the congestion increment; (14) bounds W_q from above and so cannot supply that floor by itself. By Lemma 1 the wait is strictly increasing once $M > c$, hence $\delta \triangleq \Delta W_q(c+1, c) > 0$, and by Assumption 2 the increments are non-decreasing, so $\Delta W_q(M) \geq \delta$ for every $M \geq c+1$. On the coverage side Assumption 1 gives $c_1(M) = \Theta(1/M)$, whence $|\Delta c_1(M)| = \Theta(1/M^2) \rightarrow 0$, and there is a finite $M_1 \geq c+1$ with $|\Delta c_1(M)| < \delta$ for all $M \geq M_1$. For those M , $\Delta \text{AoI}(M) \geq \delta - |\Delta c_1(M)| > 0$: AoI is strictly rising, and no minimizer lies beyond M_1 , so $\mathcal{M}^*$ is finite.

A non-decreasing difference sequence that is negative for small M and strictly positive for every $M \geq M_1$ changes sign exactly once. Let M^* be the first index at which ΔAoI stops being negative. Then AoI is strictly decreasing on $\{1, \dots, M^*\}$ and non-decreasing afterwards, so $\mathcal{M}^*$ consists of M^* together with the run of consecutive indices immediately above it on which ΔAoI vanishes. That set is a block of consecutive integers, and it collapses to the singleton $\{M^*\}$ exactly when no forward difference is zero. This establishes the U-shape. We note that discrete convexity alone does *not* force uniqueness: a single vanishing forward difference at the crossing produces two minimizers of equal value, which is why the statement is phrased in terms of $\mathcal{M}^*$.

For the provisioning law, fix M and increase c . The same monotone coupling, now applied to the down-rates $\min(n, c)\mu$ (which increase with c), makes the stationary occupancy stochastically smaller, so $W_q(M, c)$ is decreasing in c (more ports serve the same population); by Assumption 2 its increments $\Delta W_q(M; c)$ are non-increasing in c as well. Enlarging the station thus both lowers the wait and defers the onset of the steep saturation region to a larger population. The coverage difference $\Delta c_1(M)$ is unchanged. Hence for $c' > c$ we have $\Delta \text{AoI}(M; c') \leq \Delta \text{AoI}(M; c)$ for all M , so the single sign change of $\Delta \text{AoI}(\cdot; c')$ occurs at an index at least as large as that of $\Delta \text{AoI}(\cdot; c)$, i.e. $M^*(c') \geq M^*(c)$. ■

Is the tie a real risk, and does the data support Assumption 2? Both questions are settled by measurement rather than by assertion, on the physical AoI curves of Section VI. Across the 7 port capacities we simulate, the minimizer is unique in all 7 cases, and the closest competing swarm size sits at least 0.20% above the minimum, so no configuration comes near the exact

cancellation that would enlarge $\mathcal{M}^*$. In 5 of the 7 cases the minimizer is interior; for the two largest capacities the curve is still falling at the largest swarm size we simulate, so the threshold lies beyond the tested range and we do not claim to have observed it there. For Assumption 2, 32 of the 126 second differences of the simulated curves are negative, but 0 of them exceed twice the Monte-Carlo standard error of the differences involved. The observed departures from convexity are therefore consistent with simulation noise, and we found no configuration in which convexity fails by a margin the simulation can resolve.

Proposition 1 says that congestion is a genuine, capacity-limited effect rather than an artifact of poor routing: even the balanced single-cell wait of Lemma 1 forces a finite M^* . The provisioning law $M^*(c)$ non-decreasing is the design counterpart: to operate a larger swarm at low AoI, one must add ports, not merely UAVs. Both the U-shape and the rightward shift of M^* with capacity are confirmed empirically, under optimal (load balancing) assignment, in Section VI.

B. Placement inversion under contention

For a single UAV the wait term in (13) vanishes ($M = 1 \leq c$), so $\text{AoI} = c_1(1) + 2 \text{dist}/V + \tau_{\text{charge}}$ depends on placement only through the recharge detour dist . The AoI-optimal station therefore sits where the detour is smallest, i.e. near the trajectory centroid: placement is coverage-driven. The next result shows this rule is not merely loose but can be strictly wrong for a swarm.

Proposition 2 (Placement inversion): There exist instances in which the coverage-optimal (minimum-detour) placement is strictly AoI-suboptimal for a swarm. Concretely, with a per-station port cap and two candidate sites, opening only the nearest site is beaten by opening both sites and splitting the swarm, for every swarm size above a finite threshold M_0 .

Proof: Consider two candidate sites and a per-station port cap $C_{\text{port}} = c$ (Section III), so a single station can host at most c ports; adding capacity beyond c requires opening a second station. Site 1 is central with zero extra detour; site 2 is offset, adding a round-trip detour $\delta = 2\Delta/V > 0$ for any UAV assigned to it. Two placements are compared under the same swarm and field, so the coverage term $c_1(M) = t_{\text{loop}}$ is identical in both:

$$(A) \text{ coverage-optimal: } \text{AoI}_A(M) = t_{\text{loop}} + W_q(M, c) + c_0, \quad (17)$$

$$(B) \text{ split: } \text{AoI}_B(M) = t_{\text{loop}} + \delta + W_q\left(\frac{M}{2}, c\right) + c_0, \quad (18)$$

where (A) sends all M UAVs to the near station (its c ports serving population M), and (B) opens both stations with c ports each and assigns $M/2$ UAVs to each; the peak in (B) is attained by the UAVs on the offset station, which carry the extra detour δ . Subtracting,

$$\text{AoI}_A(M) - \text{AoI}_B(M) = \left[W_q(M, c) - W_q\left(\frac{M}{2}, c\right) \right] - \delta. \quad (19)$$

Two bounds make the bracket diverge. For the near station, the throughput cannot exceed the service ceiling, $\lambda_{\text{eff}} \leq$

$c\mu$; combined with the finite-source cycle identity $\lambda_{\text{eff}} = M/(T_{\text{up}} + W_q + \tau_{\text{charge}})$ (each of the M UAVs alternates operating, waiting, and charging), this gives the linear lower bound

$$W_q(M, c) \geq \frac{M}{c\mu} - T_{\text{up}} - \tau_{\text{charge}}. \quad (20)$$

For the split station, the Lemma upper bound (14) gives $W_q(M/2, c) \leq (M/2 - c)/(c\mu) \leq M/(2c\mu)$. Subtracting,

$$W_q(M, c) - W_q\left(\frac{M}{2}, c\right) \geq \frac{M}{2c\mu} - T_{\text{up}} - \tau_{\text{charge}}, \quad (21)$$

which grows without bound in M (the constants T_{up} and τ_{charge} do not depend on M), whereas δ is fixed. Therefore there is a finite threshold

$$M_0 = \min \left\{ M : W_q(M, c) - W_q\left(\frac{M}{2}, c\right) > \delta \right\} \quad (22)$$

such that $\text{AoI}_A(M) > \text{AoI}_B(M)$ for all $M > M_0$. For such M the minimum-detour placement (A) is strictly AoI-suboptimal, so the AoI-optimal placement must open the offset site and balance the charging load, even at the cost of a longer detour. ■

What this proposition does and does not claim. The mechanism above is the per-station cap: opening the second site is what makes the extra ports available, so configurations (A) and (B) differ in total capacity as well as in geometry. The proposition is therefore an existence result about placement *under a cap*, and it should not be read as separating a siting effect from a capacity effect. That separation is an empirical question, and Section VI-E answers it by measurement: at a fixed total port budget, and on the subset of instances where the two strategies buy an identical allocation, traffic-driven siting still wins. We have no proof of the matched-capacity statement and do not claim one.

Proposition 2 formalises the inversion: as the swarm grows, the optimal placement stops chasing the shortest detour and starts spreading stations to balance queueing load, so the objective flips from coverage-driven to traffic-driven. The single-UAV placement rule of the base model is thus not a conservative approximation but the wrong optimizer for a contended swarm. This crossover, including the emergence of the load-balanced placement as the field and swarm scale, is demonstrated empirically in Section VI.

V. JOINT SOLVER

Problem (P0) in (12) couples binary placement, integer port capacity, continuous trajectories, a combinatorial assignment, and the nonconvex finite-source wait (10). Rather than attack this MINLP monolithically, we decompose it into blocks that are each tractable and solve them in a loop: a trajectory block, a finite-source queue evaluation, a port-allocation block, an assignment block, and an outer placement search. Fig. 2 sketches the flow. The blocks share the peak-AoI objective (11), so a change accepted by one block is scored by the same criterion the others use: the assignment search drives the peak AoI down monotonically to a local optimum, while the placement and port searches are enumerated exhaustively and retain the best configuration found. We describe each block, then the overall algorithm.

A. Trajectory block (GPU CETSP)

A UAV collects from any sensor within a horizontal radius r_c , so a patrol need not visit every sensor point: it need only pass within r_c of each assigned sensor. The shortest patrol is therefore the shortest closed tour of hover points whose r_c -disks cover the UAV's sub-field, a Close-Enough Traveling Salesman Problem (CETSP) [24]. We solve it by differentiable optimization. For UAV m , let $\mathbf{p}_{m,1}, \dots, \mathbf{p}_{m,n_m}$ be its hover points, initialized at the k -means centroids of the sub-field. Within a cyclic order the closed-tour length is differentiable in the coordinates, so we minimize it by gradient descent. To also optimize the visiting order, and not freeze it at initialization, we alternate the continuous and combinatorial parts: we run a block of gradient steps on the coordinates, then reorder the current hover points by a 2-opt improvement of the tour, then repeat. This is a coordinate descent between the continuous positions and the discrete sequence, so the order adjusts as the positions move. The objective is the total closed-tour length plus a coverage penalty,

$$\mathcal{L} = \sum_{m \in \mathcal{M}} \left[\underbrace{\sum_{j=1}^{n_m} \|\mathbf{p}_{m,j+1} - \mathbf{p}_{m,j}\|}_{\text{tour length}} + \underbrace{\lambda_{\text{cov}} \sum_{k \in \mathcal{K}_m} [d_{m,k}^{\min} - r_c]_+^2}_{\text{coverage violation}} \right], \quad (23)$$

where indices are cyclic ($\mathbf{p}_{m,n_m+1} \equiv \mathbf{p}_{m,1}$), $\mathcal{K}_m$ is the sensor set assigned to UAV m , $d_{m,k}^{\min}$ is the distance from sensor k to its nearest hover point, $[\cdot]_+ = \max(0, \cdot)$, and λ_{cov} weights coverage against length. We minimize (23) by Adam over the hover-point coordinates, interleaved with the 2-opt reordering above. All UAVs of a scenario are stacked into one padded, masked batch and the gradient steps run in parallel on a GPU, so the whole swarm (and, by stacking seeds, whole sweeps) is optimized together. The block returns each UAV's optimized tour length, hence its flight time $t_{\text{fly}}^{\text{loop}}$ and, through (4), its revisit period t_{loop} and flight budget τ_{fly} . Because the sub-fields are fixed by the field partition, this block is independent of placement and is solved once at the start.

B. Charging queue and port allocation

Given an assignment, each open station s is the finite-source queue of Section III. Its mean wait $W_q(N_s, c_s)$ follows the machine-repair balance equations (8) to (10) from the number of assigned UAVs N_s , the ports c_s , the operating time (6), and the charge time τ_{charge} ; this evaluation is what turns a candidate configuration into a peak AoI through (11). Ports are then allocated across the open stations to reduce the peak wait under the budget $\sum_s c_s \leq C_{\text{tot}}$. Since W_q is convex-decreasing in the port count, the peak-reducing allocation adds ports where they buy the most, that is, to the most contended stations: a water-filling allocation. We realize it by scanning the integer port splits that sum to C_{tot} and keeping the split that yields the lowest peak AoI, which is exact for the small station budgets S_{max} considered here.

C. Assignment

With placement and ports fixed, UAVs are assigned to open stations to balance the queueing load. Starting from the nearest-station assignment, we run a greedy load-balancing local search: scanning the UAVs in turn, we apply the first single-UAV reassignment that strictly reduces the peak AoI (11), restart the scan after each accepted move, and stop when no single reassignment improves it. Because the peak AoI is set by the worst station, driving it down repeatedly pulls UAVs off the most contended station onto lighter ones, so the search approximates the min-max (load-balancing) assignment $\arg \min_a \max_s W_q(N_s, c_s)$ without enumerating all assignments.

D. Placement (outer search)

The open-site subset $\mathbf{x}$ is chosen by an outer search over the candidate combinations. Candidate sites are the interior nodes of a regular grid over the field, and the search enumerates the $\binom{S}{S_{\max}}$ subsets of size exactly $S_{\max}$. Two reviewers asked why it does not also consider *fewer* than $S_{\max}$ stations, which the budget constraint permits. It does, implicitly: a subset that opens $S_{\max}$ sites but gives zero ports to one of them is evaluated inside the port-split enumeration and *is* a configuration with fewer open stations, since a portless site can take no UAV. Enumerating subsets of exactly $S_{\max}$ sites therefore loses no solution; it only changes how the same set is indexed. For each subset, the inner blocks (port allocation, assignment, and the queue evaluation) are run and the resulting peak AoI recorded; the subset with the lowest peak AoI is kept. This outer search is where the placement flips from coverage-driven to traffic-driven: a subset that spreads stations to split the charging load can beat the minimum-detour subset once contention is priced in through W_q .

E. Overall algorithm

Algorithm 1 assembles the blocks. The trajectory block is solved once, after which the placement, port, and assignment blocks are alternated: for each candidate placement the port split and the load-balancing assignment are optimized against the same peak-AoI objective, each evaluated through the finite-source queue.

What Algorithm 1 relaxes. Reviewers asked us to state precisely which constraints of (P0) the solver enforces and which it does not, so we list them. Enforced at every candidate: the station and port budgets (12e), (12f) by construction of the enumeration; assignment to open stations (12h) by construction; and the recharge-reachability constraint (12c), which is re-checked after *every* reassignment because the peak-AoI evaluation returns $+\infty$ when any UAV would exceed its one-way energy budget, and the greedy local search calls that evaluation on each candidate move. Section VI-I reports the resulting margin.

Relaxed: the inter-UAV separation constraint (12i). It is not enforced, and it cannot be enforced at this level of abstraction, because the planner works in cycle times ($t_{\text{loop}}, \tau_{\text{fly}}$) and does not resolve where each UAV is at a given instant; two UAVs on

Algorithm 1 Block-coordinate joint solver for (P0)

Input: sensors $\mathcal{K}$, candidate sites $\mathcal{S}$, station budget $S_{\max}$, port budget C_{tot} , swarm size M
Output: placement $\mathbf{x}$, ports $\mathbf{c}$, assignment $\mathbf{a}$, trajectories $\{\mathbf{q}_m\}$, peak AoI $\bar{A}$

- 1: partition $\mathcal{K}$ into M sub-fields
- 2: **trajectory block:** minimize (23) by Adam on the GPU for all UAVs; obtain $t_{\text{loop},m}, \tau_{\text{fly},m}$
- 3: $\bar{A}^* \leftarrow +\infty$
- 4: **for** each site subset $\mathbf{x}$ of size $S_{\max}$ in $\mathcal{S}$ **do**
- 5: **for** each port split $\mathbf{c}$ with $\sum_s c_s = C_{\text{tot}}$ **do**
- 6: $\mathbf{a} \leftarrow$ nearest-station assignment
- 7: **repeat**
- 8: scan single-UAV reassignments; apply the first one that strictly reduces the peak AoI (11) (evaluating W_q by (10)) and restart the scan
- 9: **until** no single reassignment reduces the peak AoI
- 10: $\bar{A} \leftarrow$ peak AoI of $(\mathbf{x}, \mathbf{c}, \mathbf{a})$
- 11: **if** $\bar{A} < \bar{A}^*$ **then**
- 12: $\bar{A}^* \leftarrow \bar{A}$; store $(\mathbf{x}, \mathbf{c}, \mathbf{a})$
- 13: **end if**
- 14: **end for**
- 15: **end for**
- 16: **return** stored $(\mathbf{x}, \mathbf{c}, \mathbf{a}), \{\mathbf{q}_m\}, \bar{A}^*$

adjacent sub-fields could in principle be close at some moment that the model does not represent. Algorithm 1 is therefore a heuristic for a *relaxed* problem, not a solver for (P0) as written. What we can do is measure the cost of the relaxation, and we do: over 20 seeds and 1320 UAV pairs, the closest approach between the hover points of two different UAVs is 199 m, with a median pair separation of 2227 m; 0 pairs fall below 100 m and 1 falls below 200 m. The returned tracks are thus spatially well separated even though nothing in the solver forces them to be. This is a spatial statement about patrol tracks, not a time-resolved collision guarantee; a deployment would still delegate separation to a lower-layer avoidance controller or to altitude stratification.

Complexity and convergence. The trajectory block costs one batched GPU optimization independent of the placement search. The outer loops evaluate $\binom{S}{S_{\max}}$ placements times the number of port splits, and each evaluation runs the greedy assignment, whose local search terminates because every accepted move strictly lowers the peak AoI (a quantity bounded below by zero) and there are finitely many assignments. For a fixed placement and port split, the returned assignment is a local optimum of (11) under single-UAV moves; because the placement and port searches are exhaustive over the candidate grid and the integer splits, and the best configuration is retained, the solver returns the lowest peak AoI it evaluates. We do not claim global optimality: the hover-point positions and their visiting order are optimized only to a local optimum of (23), and the assignment search is greedy, so the result is the best configuration reachable by this search from the nearest-station initialization, not necessarily the global optimum of (P0).

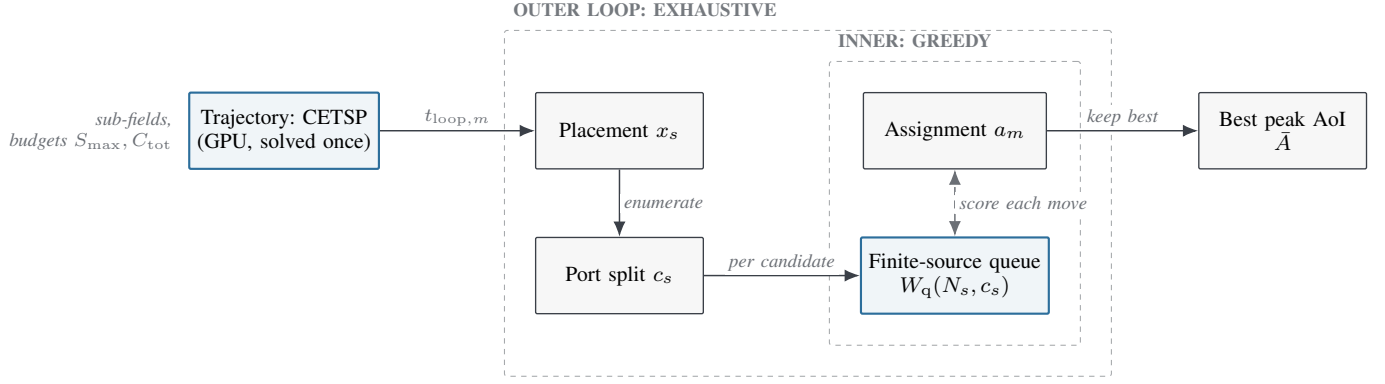


Fig. 2. Block-coordinate solver. The trajectory is solved once, before the placement search, so the sub-fields and patrol tours are fixed for every candidate; Section VI-G measures what that costs. The placement and port-split loops are enumerated exhaustively over the candidate grid, whereas the UAV-to-station assignment inside them is a greedy local search whose every move is scored by the finite-source queue. The queue is therefore the objective of the inner search, not a stage of a pipeline. Blue marks the two blocks this paper contributes.

VI. EXPERIMENTS

A. Setup

We simulate $K=60$ sensors uniformly placed in a square field (side length varied from 5 to 20 km), a swarm of M rotary-wing UAVs cruising at $V=12$ m/s at altitude $H=100$ m, and a battery of about 67 Wh with a 15% reserve. The charge time per port is $\tau_{\text{charge}}=8$ min. The propulsion power follows the rotary-wing model [43], and the air-to-ground link uses a line-of-sight free-space rate. Candidate stations lie on a 3×3 grid; unless stated otherwise the budget is $S_{\text{max}}=2$ open stations and $C_{\text{tot}}=4$ ports. In the joint comparison we use a swarm of $M=12$ UAVs. A reviewer asked us to re-examine that choice, and the re-examination changed what we can claim. Table IV gives $M^*=12$ for a *per-station* capacity of $c=4$, i.e. for one station with four ports; the comparison setup instead splits $C_{\text{tot}}=4$ over up to two stations, which is not the same queue. Sweeping M directly under the actual placement and port split at the comparison field size (Section VI-M) gives $M^*=16$, not 12. We keep $M=12$, which costs the proposed method 3.9% relative to its own optimum, for two reasons: every method peaks at the *same* $M^*=16$, so the choice favours none of them, and Section VI-M repeats the whole comparison with each method at its own optimum and finds the ranking and the magnitudes essentially unchanged. We therefore no longer describe $M=12$ as the AoI-optimal swarm size, only as a fixed and neutral operating point. The scaling study in Section VI-K covers genuinely large swarms, up to $M=60$, by growing the charging capacity with the swarm. Every reported value is a mean over 20 to 30 random sensor fields (fixed seeds), with the standard deviation shown where relevant.

Implementation, hyperparameters and runtime: All results come from a *single* machine, so no reported quantity mixes hardware: a NVIDIA GeForce RTX 4090 with 80 CPU cores, PyTorch 2.6.0+cu124 on CUDA 12.4, Python 3.11.16. The trajectory block minimizes (23) with Adam at learning rate 8.0 and coverage weight $\lambda_{\text{cov}}=50.0$: 400 steps on the hover-point positions, then 3 rounds that each reorder the visiting sequence by nearest-neighbour plus 2-opt and polish it with $\lfloor 400/3 \rfloor$ more steps, 799 gradient steps in all. The budget

is *fixed*: no early-stopping threshold and no random restarts, so a run is determined by its seed. Each value averages 20 seeds.

One solve takes 3.09 s at $M=12$ and 3.50 s at $M=60$, and the split matters more than the total because the blocks scale in opposite directions. The GPU trajectory block costs 3.03 s and 2.78 s, barely growing with the swarm, since all UAVs are optimized in one padded batch and a larger swarm means smaller sub-fields with fewer hover points each; the CPU combinatorial block instead grows by almost an order of magnitude, 0.07 s to 0.69 s, as the assignment search is re-scored per UAV. The GPU thus accounts for 98% of a solve at $M=12$ but 80% at $M=60$. Timing discards a CUDA warm-up call, synchronizes the device around each timer, and is validated by requiring the two block times to sum to an independently timed solve.

Table III lists every constant, including the rotary-wing power coefficients and the link budget; it is generated by importing the values directly from the artifact source, so it cannot drift from the code that produced the results.

B. Queue-model validation against a discrete-event simulation

The whole analysis rests on the charging-queue model, so we first validate it. We built a discrete-event simulation (DES) of the actual cycling UAVs, with a first-in-first-served station and c ports, in which each UAV alternates between patrolling for a randomized operating time and traveling to the station to recharge. The simulation supports both exponential and near-deterministic operating and service times, and we run each configuration over eight seeds and a long horizon after a warm-up, then measure the true mean queue wait. Fig. 3 compares three predictions against the DES. Under matched assumptions (exponential operating and service), the finite-source $M/M/c/N$ formula matches the DES to a mean relative error of 4.2% over the 25 configurations in which the DES wait is resolvable, which confirms the formula. For the realistic system (near-deterministic operating and charging), the DES wait is lower, so the finite-source model is a *conservative upper bound* (it exceeds the real wait in 23/23

TABLE III

SIMULATION PARAMETERS. ALL VALUES ARE IMPORTED DIRECTLY FROM THE ARTIFACT SOURCE BY THE SCRIPT THAT GENERATES THIS TABLE, SO THE TABLE CANNOT DRIFT FROM THE CODE THAT PRODUCED THE RESULTS.

Parameter	Value	Unit
<i>Field and sensors</i>		
Sensors K	60	–
Field side L	5 to 20	km
Collection radius r_c	200	m
Payload per visit	5	Mb
<i>UAV and flight</i>		
Cruise speed V	12	m/s
Altitude H	100	m
Battery $E_{\max}$	66.7	Wh
Energy reserve	15	%
<i>Rotary-wing power model [43]</i>		
Blade profile P_0	79.86	W
Induced power P_i	88.63	W
Tip speed U_{tip}	120	m/s
Hover induced velocity v_0	4.03	m/s
Fuselage drag ratio d_0	0.60	–
Air density ρ	1.225	kg/m ³
Rotor solidity s	0.050	–
Rotor disc area A	0.503	m ²
<i>Air-to-ground link</i>		
Reference gain β_0	-50	dB
Sensor transmit power p_{tx}	20	dBm
Noise PSD N_0	-170	dBm/Hz
Bandwidth B	1	MHz
<i>Charging and placement</i>		
Charge power per port	500	W
Charge time τ_{charge}	8.0	min
Candidate sites	3 × 3 grid	–
Open stations $S_{\max}$	2	–
Total ports C_{tot}	4	–

configurations). In contrast, the open $M/M/c$ queue overpredicts the DES wait by a mean factor of 5.6, because it ignores the closed population: only M UAVs exist, so the arrival rate falls as more of them are already charging. Using the open queue therefore distorts the design decisions (we show below that it produces a spurious placement crossover), which is why we adopt the finite-source model throughout. The finite-source $M/M/c//N$ model assumes exponential service and is thus a conservative bound: it exceeds the real wait by a mean factor of 4.5. Replacing the exponential service by a deterministic-service $M/D/c//N$ approximation, obtained with the standard service-variability correction (the wait scales with $(1+c_s^2)/2$, where c_s is the squared coefficient of variation of the service time, so $c_s^2=0$ halves the wait), tightens this to $2.3\times$ (Fig. 3). The residual gap is due to the near-deterministic operating process, which a service-only correction does not capture; since both models are upper bounds, the reported AoI values remain safe rather than optimistic.

C. Proposition 1: congestion threshold and provisioning

Fig. 4 plots the peak AoI against swarm size M for several port budgets. Each curve is U-shaped: increasing M first lowers AoI (denser coverage) and then raises it (charging congestion), confirming the congestion threshold of Proposition 1. The optimal swarm size M^* grows with the charging capacity,

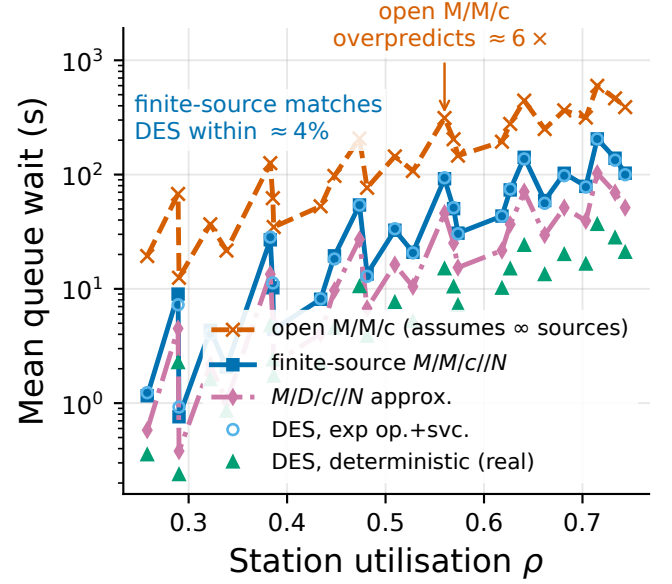


Fig. 3. Queue-model validation against discrete-event simulation. The finite-source model matches the DES under matched (exponential) assumptions and is a conservative upper bound for the real near-deterministic system, whereas the open $M/M/c$ overpredicts the wait about $6\times$.

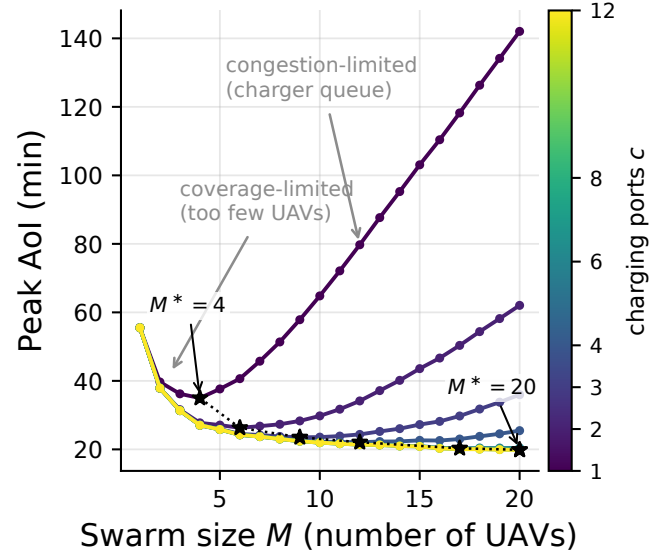


Fig. 4. Peak AoI is non-monotone in the swarm size M (finite-source model). Each port budget has an interior optimal swarm size M^* (black markers).

as summarized in Table IV, which is exactly the provisioning law: more ports support a larger cost-effective swarm.

D. Proposition 2: placement inversion

We compare, per instance, the coverage-optimal placement (minimum recharge detour, contention-blind) against the traffic-optimal placement (which balances the queueing load). The traffic-optimal placement lowers the peak AoI on 19/20 seeds and ties on the remaining 1; it never increases it. The mean reduction is 14.8% (median 17.2%), and the traffic-

TABLE IV

PROVISIONING LAW: OPTIMAL SWARM SIZE M^* VERSUS PORT BUDGET c . ENTRIES MARKED $\geq$ SIT AT THE CEILING OF THE SWEEP SWARM SIZE ($M \leq 20$): THEY ARE *censored* EXTREMA AND MUST NOT BE READ AS INTERIOR OPTIMA.

Ports c	1	2	3	4	6	8	12
M^*	4	6	9	12	17	≥ 20	≥ 20

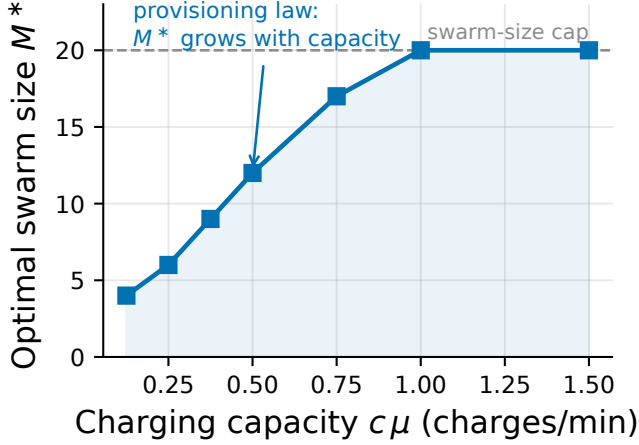


Fig. 5. Provisioning law: the optimal swarm size M^* increases with the charging capacity, so a larger port budget supports a larger cost-effective swarm.

optimal solution selects *different* station sites on 18/20 seeds (Fig. 6).

We are careful about what this does and does not establish. Proposition 2 is an existence result whose mechanism is a *per-station* port cap: because a single site cannot hold more than c ports, buying capacity requires opening a second site, and the proof therefore compares configurations with different total capacity. The experiment above holds the total port budget fixed and is thus not a test of that mechanism. It is a separate and, in this respect, stronger empirical claim: the inversion appears even when no extra capacity is bought. Section VI-E isolates it further. Neither result subsumes the other, and we do not present the measurement as a verification of the proposition.

E. Separating placement from installed capacity

A reviewer observed that the analytical comparison behind Proposition 2 contrasts one station holding c ports with two stations holding c ports each, so that the second configuration enjoys both a different placement and twice the installed capacity. We agree that the two effects must be separated, and we do so here rather than by argument.

The experiment of Section VI-D already holds the *total* port budget fixed: both strategies receive the same C_{tot} and differ only in how they distribute it. Across 20 seeds the traffic-driven placement lowers peak AoI by 14.8% on average and is at least as good on 19 of 20 seeds. That comparison still allows the two strategies to allocate ports differently between sites, so we split it further. On the 12 seeds where the two strategies

Traffic-driven placement lowers peak AoI mean –14.8%, sites differ on 18/20

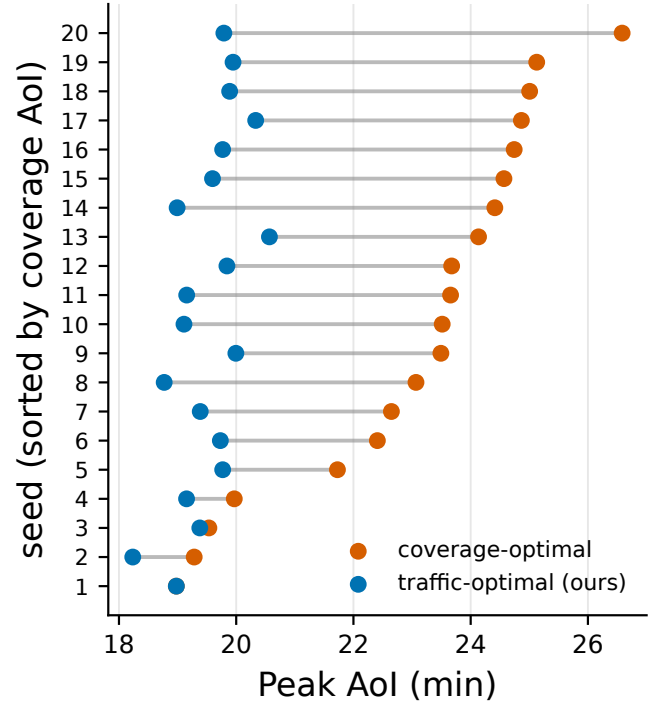


Fig. 6. Placement inversion. Traffic-optimal placement (which balances the queueing load) is never worse than the coverage-optimal (minimum-detour) placement: it wins strictly on 19 of 20 seeds and ties on the remaining 1. It chooses different sites on 18/20 seeds.

choose *identical* port allocations, and therefore differ only in *where* the stations stand, the traffic-driven choice is still better by 16.1% on average, with a median of 18.6%, and wins on 11 of 12. On the remaining 8 seeds, where allocations do differ, the mean gain is 12.7%.

The effect therefore does not come from added capacity: it is, if anything, larger when capacity is held identical in every respect.

Two caveats keep this honest. First, this is a measurement over 20 instances of one scenario family, not a theorem: we have no proof that traffic-driven siting dominates at matched capacity in general, and Proposition 2 does not supply one, because its argument runs through the per-station cap. Second, we retain the cap in the statement of Proposition 2 rather than restating the proposition at matched capacity, because the cap is what makes the analytical comparison tractable; the matched-capacity claim is empirical and is stated as such.

F. Does the queue approximation change the ranking?

Section VI-B reports that the finite-source model is conservative and that the gap widens under near-deterministic operation. The question that matters for the design recommendation is not the size of that gap but whether it reorders the candidates. We therefore repeat the placement comparison twice on the same seeds, once with the exponential-service finite-source wait used throughout the paper and once with the

deterministic-service correction of Section VI-B, and compare the resulting *orderings*.

The coverage-driven placement is better on 0 of 20 seeds under either model. On 2 seeds the two placements are exactly tied, and on 1 seed the deterministic-service correction reduces a narrow advantage to an exact tie. No seed reverses the ordering. The reported gain is likewise stable: within this single run, replacing the exponential-service model by the deterministic-service correction changes the mean placement gain by 0.7 percentage points. The recommendation and its magnitude are thus insensitive to the service-time assumption over the range we can check, even though the absolute waiting times are not.

That test still scores every design with an *analytic* wait, so it cannot rule out a bias common to all of them. We therefore ran the stronger check a reviewer asked for: one *common* discrete-event simulator, with near-deterministic operating and charging, applied to the *final design* of all 5 methods. For each returned configuration we simulate every open station with its own assigned UAVs and port count, substitute the simulated wait for the analytic one in (11), and leave everything else unchanged; as a control, repeating the substitution with the analytic wait reproduces the solver's own value to machine precision, so the simulator is the only thing that differs. The analytic model is conservative for every method, by 0.6 to 3.5%, and the *ordering is identical* under the two evaluators. Under the common simulator the proposed design reaches 40.4 min against 45.9 min for the single pooled station, a gain of 12.0%, and it is the best design on 20 of 20 seeds. The reported ranking is therefore not an artefact of the analytic surrogate.

G. What the block decomposition costs

The formulation in (12) is joint in placement, capacity, trajectory and assignment. The solver is not: it partitions the field, solves the trajectory block once, and then searches over placements and assignments. Since a station's position changes the recharge detour, that ordering is an approximation and its cost should be measured rather than asserted.

We measure it with one round of limited joint re-optimisation. Holding the sub-fields and their patrol tours fixed, we reassign UAVs to sub-fields once the stations are known, using an optimal assignment on the resulting detours, and then re-solve the placement block. Averaged over 20 seeds the second round changes peak AoI by 0.00%, that is, it is indistinguishable from no change: it helps on 5 seeds, is slightly worse on 3, and its largest single-seed gain is 2.31%. The ordering is therefore an approximation whose cost we measure rather than assert, and that cost is negligible here.

H. Why the objective is stated as an expectation

Section III-E defines the objective as the expected peak age, that is, the average over cycles of the age reached just before each delivery. Two alternatives, a sample-path worst case and a time-average age, are ruled out by measurement rather than preference; we report both here.

a) *Expected peak against time-average*: For a sensor revisited every T seconds the age rises linearly from zero to T , so the expected peak is $\mathbb{E}[T]$ while the time-average is $\mathbb{E}[T^2]/(2\mathbb{E}[T])$. The two therefore differ by a factor $2/(1 + CV^2)$. Building the cycle-length distribution from the simulated wait samples, the measured ratio is 2.00 to 2.00 across 9 configurations, since the cycle length varies little (coefficient of variation at most 0.040). The two names are not interchangeable: a time-average would halve every AoI value here.

b) *Expected peak against sample-path worst case*: Using the discrete-event simulator of Section VI-B under near-deterministic operation and charging, we record the full distribution of station waiting times rather than its mean alone. Across the load points we examine, the 95th percentile of the wait exceeds the mean by a median factor of 1.40, and the largest observed wait exceeds it by a median factor of 4.89. The spread is strongly regime dependent: where the queue is saturated the mean is representative, with a maximum-to-mean ratio of 1.11, whereas in the lightly loaded configurations the same ratio reaches 82.9, because most excursions wait not at all and a few wait a full service time.

The consequence is specific: a mean-based expression is an adequate surrogate for a worst case exactly where the system is congested, and inadequate exactly where it is well provisioned, the operating point Section VI-O recommends. We therefore state the objective as an expectation throughout, and we do not claim worst-case guarantees for the placements we select.

I. Feasibility, and where it stops holding

Constraint (12c) requires every UAV to keep enough energy to reach its station. The solver re-evaluates it at every candidate, not once: the AoI evaluation returns $+\infty$ when a UAV would exceed its one-way budget and the assignment refinement calls it after each swap, so an infeasible configuration is never returned *when the constraint is active*. That qualification matters, and a reviewer's request to report every constraint violation is what exposed it. The one-way budget is $E_{\text{res}}E_{\text{max}}/P(V) = 282$ s of flight, i.e. 3.38 km. With $S_{\text{max}}=2$ stations on the candidate grid, that budget is enough to serve the whole swarm only in small fields. Table V sweeps the field size: the constraint is satisfied on all 20 seeds up to $L=6$ km, starts to fail at 7 km, and admits no feasible configuration at all from 9 km onward. The comparison field of 15 km is past that point, so the experiments there, and the field sweep beyond 6 km, are run with (12c) *relaxed*; the largest one-way overshoot at 15 km is 806 s of flight. At $L \leq 6$ km the smallest slack anywhere is 7.7 s with 0 infeasible configurations, but that figure does not describe the large-field results, so we do not let it stand for the whole study. Two things follow. The large-field AoI values hold *under the assumption that a UAV can always reach its station* and should be read that way; and the shortfall is a property of the infrastructure, not of the placement algorithm: opening a third station restores feasibility up to 9 km and a fourth up to 12 km, but from 15 km no subset of the 3×3 candidate grid suffices, because a corner sub-field is farther than 3.38 km from every candidate

TABLE V

WHERE THE RECHARGE-REACHABILITY CONSTRAINT (12c) CAN BE MET, FOR $S_{\max}=2$ STATIONS AND A ONE-WAY BUDGET OF 3.38 KM. “STATIONS NEEDED” IS THE SMALLEST NUMBER OF OPEN STATIONS THAT RESTORES FEASIBILITY.

Field L (km)	Feasible seeds	Max overshoot (s)	Stations needed
5	20/20	0	–
6	20/20	0	–
7	17/20	226	3
8	3/20	298	3
9	0/20	371	3
10	0/20	443	4
12	0/20	588	4
15	0/20	806	6
20	0/20	1168	6

site. Serving such fields needs a denser candidate grid, a larger reserve, or in-field charging, a provisioning statement Section VI-O now carries. Every other constraint of (P0) holds exactly on every returned configuration: station and port budgets and open-station assignment show zero violation on all seeds, and the collection-radius penalty leaves at most 0.61 m residual. The separation constraint (12i) is relaxed throughout, as discussed in Section V.

J. Joint solver versus baselines

We compare the proposed joint solver (CETSP trajectory with traffic-optimal placement) against: a single pooled station (all ports at one central station, the strong Wei-style baseline [3]), a coverage-optimal placement, a round-robin assignment, and an ablation that replaces the CETSP trajectory with a nearest-neighbor patrol. Table VI reports the ablation at a 15 km field: the proposed method attains the lowest peak AoI, the CETSP trajectory contributes 3.9% on average (bootstrap 95% confidence interval [3.6, 4.2]%, per-seed range 2.0 to 5.4%), and the contention-aware placement contributes substantially more (about 18% over the coverage-optimal placement).

The single-UAV reference makes the premise of the paper concrete rather than rhetorical. One UAV patrolling the whole field reaches a peak AoI of 136.5 min against 41.1 min for the swarm, a reduction of 69.8% on average over 20 seeds and at least 65.8% on every one of them (0 seeds favour the single UAV). This is the gap that motivates a swarm in the first place, and it is also why the charging station becomes contended: the swarm that closes this gap is exactly the swarm that must share the ports.

We state the trajectory contribution as an effect size rather than as a p -value, and the reason is worth making explicit. The proposed solver beats the no-CETSP ablation on all 20 of 20 paired seeds, so an exact two-sided signed-rank test returns $p = 2/2^{20} = 1.9 \times 10^{-6}$. That value is the *floor* of the test at this sample size: it is what any unanimous set of 20 pairs returns, regardless of whether the differences are large or negligible. It certifies the direction of the effect and nothing about its magnitude, so the interval above carries the claim.

The same caution applies to the comparison as a whole. Table VI pits the proposed method against 5 baselines at

TABLE VI

ABLATION AT A 15 KM FIELD ($M=12$, MEAN PEAK AOI IN MINUTES).

Method	Peak AoI (min)
Single UAV ($M=1$)	136.5
Proposed (CETSP + traffic-optimal)	41.1
No CETSP (nearest-neighbor patrol)	42.8
Single pooled station [3]	46.2
Coverage-optimal placement	50.4
Round-robin assignment	56.7

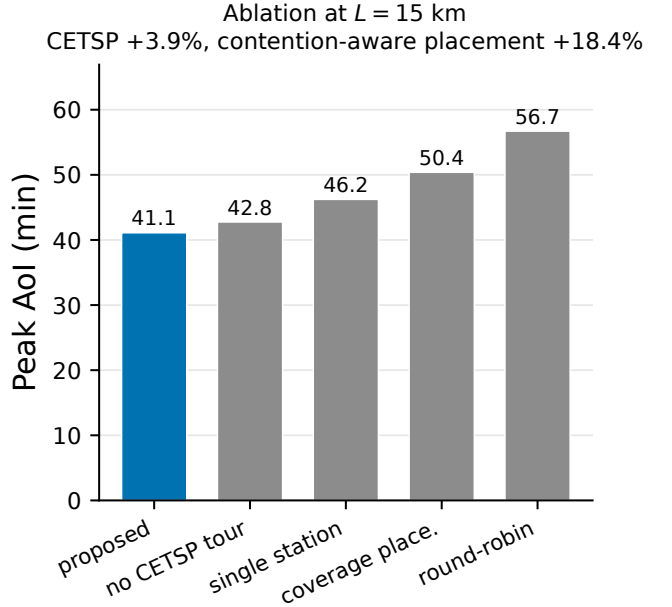


Fig. 7. Ablation at a 15 km field. Both the contention-aware placement and the CETSP trajectory contribute to the peak-AoI reduction of the proposed method. The single-UAV reference of Table VI (136.5 min) is omitted from the plot, since it is more than twice the largest bar and would compress the differences the figure is meant to show.

once, so the tests form a family and we report them as one: each is an exact two-sided paired signed-rank test, and after a Holm correction across the family every comparison remains significant at $p \leq 9.5 \times 10^{-6}$. Because unanimity pins all five raw p -values to the same floor, the informative quantities are the effect sizes, their intervals, and the *worst* seed rather than the mean. Against the single pooled station the mean gain is 10.9% (CI [9.4, 12.3]%) but only 2.4% on the worst seed; against the coverage-optimal placement, 17.6% (CI [14.1, 20.8]%) with a worst seed of 1.1%. The method wins on every seed of every comparison, but on the least favourable seed of the family the margin narrows to 1.1%, and a reader sizing a deployment should plan against that number rather than the mean.

Fig. 8 sweeps the field size. The proposed distributed, contention-aware placement beats the single pooled station at every field size, and the gain grows with the field (from 4.1% at 5 km to 12.2% at 20 km) because travel cost dominates as the field grows. The choice of queue model is decisive here. Had the charging system been modeled as an open $M/M/c$ queue, a spurious crossover would appear in which

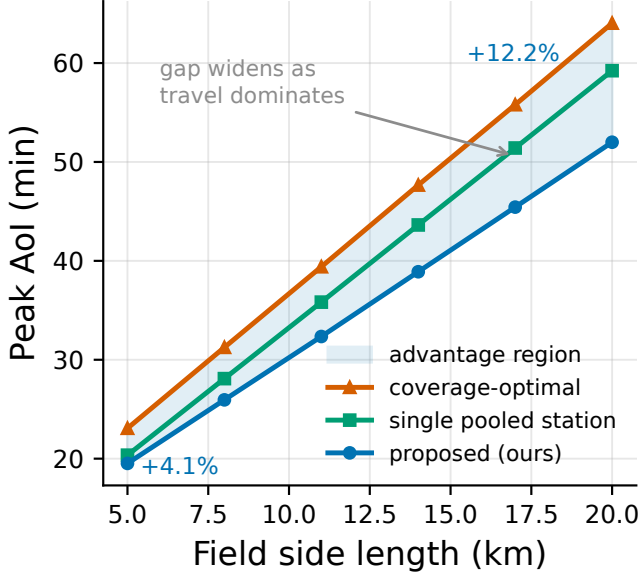


Fig. 8. Field-size sweep. The proposed distributed placement beats the single pooled station at all sizes, with a widening margin as travel cost grows.

the single pooled station wins in small fields, because the open queue overpenalizes split stations. The DES-validated finite-source model, whose wait saturates rather than diverging, removes this artifact, and the distributed placement is preferred throughout.

K. Scaling to a large swarm

A swarm is only meaningful at larger scale, so we scale the fleet to $M = 60$ UAVs over a 20 km field with $K = 8M$ sensors, and we grow the charging capacity with the swarm ($C_{\text{tot}} = \lceil M/3 \rceil$ ports) as the provisioning law of Proposition 1 requires. Fig. 9 shows that the peak AoI stays controlled as the swarm grows (from 50.6 min at $M = 20$ to 47.6 min at $M = 60$), because the sensors and the charging capacity scale together, and that the proposed contention-aware placement keeps its advantage across the whole range over both baselines, by 10.2 to 12.7% against the single pooled station and by 8.7 to 13.7% against the coverage-optimal placement. We report the two ranges separately because they do not coincide: the advantage over the coverage-optimal placement dips lower, to 8.7%, at one swarm size. The design therefore applies to genuine swarms, not only to small fleets.

L. Solver optimality

The two combinatorial blocks are heuristic in different ways, and it is worth separating them. The placement and port search is *exhaustive*, but only over a finite candidate grid of sites and integer port splits; the UAV-to-station assignment is a *greedy* local search (Section V). We therefore ask how far the pair together is from what we call the *conditional combinatorial optimum*: the optimum of the placement, port-split and assignment subproblem *conditional* on the per-UAV states returned by the trajectory block. It is *not* the optimum

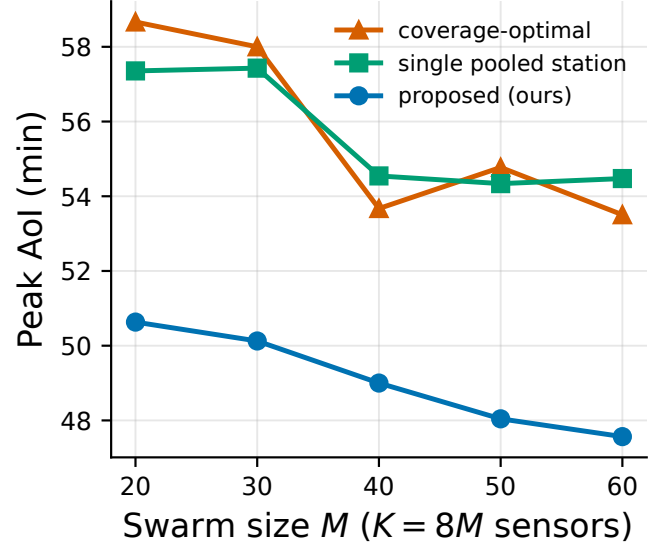


Fig. 9. Scaling to a large swarm (M up to 60, $K = 8M$ sensors, capacity $C_{\text{tot}} = \lceil M/3 \rceil$ growing with the swarm). The proposed placement keeps its advantage over the baselines at scale.

TABLE VII
OPTIMALITY GAP OF THE SOLVER VERSUS THE CONDITIONAL COMBINATORIAL OPTIMUM, I.E. THE OPTIMUM OF THE PLACEMENT, PORT-SPLIT AND ASSIGNMENT SUBPROBLEM UNDER THE PER-UAV STATES RETURNED BY THE TRAJECTORY BLOCK.

Swarm size M	3	4	5	6	7	8
Mean gap (%)	0	0	0	0	0.03	0.11
Worst gap (%)	0	0	0	0	0.51	0.99
Optimal seeds	15/15	15/15	15/15	15/15	13/15	10/15

of P0, which would also optimize the trajectories. On small instances it is computable by enumerating every open-site subset, every port split, and every UAV-to-station assignment; we run our solver and the enumeration on identical per-UAV states, so the gap measures only the combinatorial heuristic. Table VII reports the result over 15 seeds. The returned configuration is *exactly* optimal for $M \leq 6$, matching the enumerated optimum on all 15 seeds; at $M = 8$ it still matches on 10 of 15 seeds and stays within 0.11% on average and 0.99% in the worst case. The heuristic therefore matches this optimum wherever enumeration is tractable, which supports its use at larger swarm sizes; it says nothing about the gap to P0 itself.

M. Fairness of the fixed swarm size

Fixing M for every method could favour the one that chose it, so we let each method pick its own swarm size: we sweep M over a grid at the comparison field size and compare every method at *its own* optimum. All five peak at the same $M^* = 16$, so no method is disadvantaged by a common M , and no M^* sits at a grid boundary, so none of them is a censored extremum. At its own optimum the proposed method reaches 39.5 min against 41.1 min at $M = 12$. The advantages survive the change almost exactly: 11.6% over the

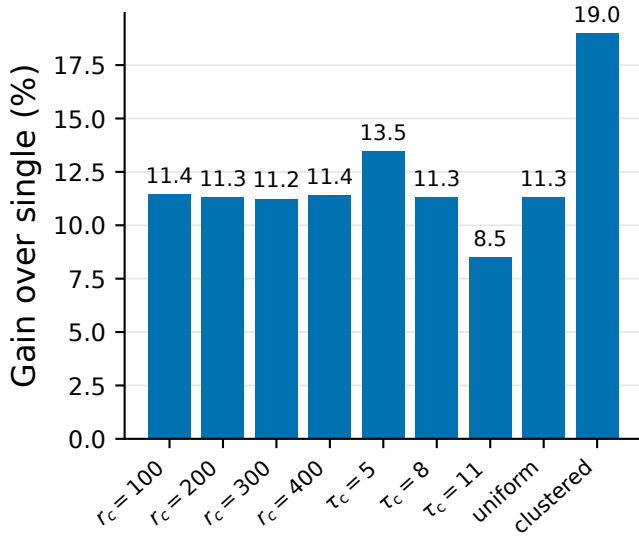


Fig. 10. Robustness of the proposed method's gain over the single pooled station to the collection radius r_c , the charge time τ_{charge} , and a uniform versus clustered sensor field ($L = 14$ km, $M = 12$).

single pooled station (versus 11.1% at $M=12$), 19.7% over the coverage-optimal placement (18.4%), 27.0% over round-robin (27.5%), and 3.8% over the no-CETSP ablation (3.9%). The fixed swarm size is therefore not what produces the reported gains.

Extreme port scarcity: The scaling study of Section VI-K grows the port budget with the swarm, holding the UAV-per-port ratio constant, so it does not test contention. Here we hold the budget at $C_{\text{tot}}=4$ ports while the swarm grows from $M=20$ to $M=60$, so the ratio rises from 5.0 to 15.0 UAVs per port. Peak AoI degrades from 55.4 min to 118.6 min, a factor of 2.14, which is the congestion regime Proposition 1 predicts and the opposite of the capacity-matched scaling. As a control we repeat the sweep with $C_{\text{tot}}=M$, one port per UAV, where contention cannot arise: peak AoI then *decreases* monotonically with M . The degradation is therefore caused by port contention, not by swarm size itself.

N. Robustness

Fig. 10 sweeps the main scenario parameters at $L = 14$ km. The gain of the proposed placement over the single pooled station is stable across the collection radius (11.2 to 11.4% for r_c from 100 to 400 m) and the charge time (8.5 to 13.5%). It is also larger, not smaller, on a non-uniform field: when the sensors form clusters the gain rises to 19.0%, because concentrated recharge traffic makes load balancing more valuable. The advantage is therefore robust and, if anything, strengthens in realistic non-uniform deployments.

O. Discussion and design guidelines

The results translate into concrete deployment guidance. First, do not size the swarm by coverage alone: the peak AoI is non-monotone in M , so beyond M^* extra UAVs only add congestion, and Table IV shows M^* is set by the installed

capacity, so fleet and ports must be provisioned together. When freshness targets tighten, adding ports is often more effective than adding UAVs to an already congested system.

Second, size the station network to the energy reserve before optimizing its placement. With a one-way budget of 3.38 km, two stations serve a field only up to $L=6$ km; three reach 9 km and four reach 12 km, while beyond 15 km no subset of a 3×3 candidate grid suffices and the grid itself must be made denser (Section VI-I). Reachability is a feasibility gate that precedes any AoI optimization.

Third, place stations to balance the queueing load, not only to minimize detour. The coverage-optimal placement concentrates recharge traffic and inflates the wait; spreading the stations lowers the peak AoI even at the cost of longer trips, and the benefit grows with the field because travel then dominates. The field-size sweep makes the exception explicit: a single pooled station remains sensible only in small fields, where travel is cheap.

Fourth, the queueing model matters as much as the optimizer: an open $M/M/c$ queue overstates the wait by about $6\times$ and can flip a design decision, as the spurious small-field crossover above shows. The finite-source model removes this bias and, being a conservative upper bound on the near-deterministic reality, yields safe rather than optimistic estimates; it is the default we recommend for swarm recharging with a bounded fleet.

Finally, the ablation isolates the gain: the contention-aware placement provides the larger share, while the GPU close-enough-TSP trajectory adds a smaller but consistent improvement by shortening the revisit period at a realistic collection radius, and the joint solver combines the two without violating the energy and budget constraints.

VII. CONCLUSION

We studied fresh data collection by a UAV swarm that recharges at a small number of shared, multi-port charging stations. We formulated the joint placement, capacity, trajectory, and assignment problem under a charging queue, and we showed that the correct queue model is a finite-source (machine-repair) queue, not an open $M/M/c$: a discrete-event simulation confirms the finite-source formula within about 4% and shows that the open queue overpredicts the wait by roughly $6\times$ and distorts the design. Two structural results followed: a congestion threshold that makes the AoI non-monotone in the swarm size, with a provisioning law linking the optimal swarm size to the charging capacity, and a placement inversion in which the AoI-optimal placement becomes traffic-driven rather than coverage-driven. A block-coordinate solver with a GPU close-enough-TSP trajectory optimizer realized these gains against strong baselines.

Several limitations remain. The finite-source model assumes exponential operating and service times; our deterministic-service $M/D/c/N$ approximation tightens the bound, but a residual gap from the near-deterministic operating process would need a full non-exponential treatment. The trajectory block is a close-enough-TSP heuristic with 2-opt reordering, not an exact solver: it matches the conditional combinatorial

optimum wherever enumeration is tractable, but guarantees at scale remain open. Finally, extending the model to stochastic data arrivals and online operation would move the design from planning toward real-time control.

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The authors used Anthropic's Claude (Opus 5) as an assistant during the preparation of this manuscript. Its use covered three areas: drafting and editing prose in the revised Introduction, Related Work, System Model, and Experiments sections; writing the analysis and experiment-orchestration scripts released with the artifact; and proposing several of the measurements reported in Section VI, including the matched-port-budget comparison, the ranking-stability check under the alternative queueing model, and the single-UAV baseline. All experimental results were produced by executing that released code, and every number reported here was checked against the run records by the authors, who take full responsibility for the content of this paper.

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